Efficient Transformer Architectures for Electrodermal Activity-based Arousal Classification

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Abstract. Electrodermal Activity (EDA) is a non-invasive biomarker of sympathetic arousal increasingly used in wearable affective computing. While Transformer-based architectures outperform convolutional baselines for subject-independent arousal classification, their computational cost limits deployment on latency-sensitive platforms. This work presents, to our knowledge, the first systematic comparison of eight efficient and modern architectures spanning five paradigms—patch-based attention, sparse attention, frequency-domain modelling, state space models, and modernised convolutions—for binary arousal classification from EDA signals under strict Leave-One-Subject-Out (LOSO) validation with 147 participants. Beyond predictive accuracy, we introduce five complementary computational efficiency metrics (parameter count, FLOPs, inference time, memory footprint, and training time) as primary evaluation dimensions, enabling explicit characterisation of the accuracy-efficiency Pareto frontier. The key finding is that Mamba—a selective state space model—achieves F1 = 0.858 at 1.5 ms inference time, approaching PatchTST accuracy (F1 = 0.863 vs. 0.858, an F1 difference that does not survive Bonferroni-Holm correction for multiple comparisons) with 3.6× lower latency and 66% fewer parameters. Channel ablation analysis further suggests that derivative channels provide greatest benefit for simpler architectures, while the additional temporal context from derivatives is partially redundant with representations learned by attention and state-space mechanisms. A classical signal processing baseline (handcrafted EDA features with SVM) achieved F1 $\approx$ 0.81 on the same dataset [26], anchoring the deep learning performance gains and confirming that architectures with global temporal modelling provide an approximately 5 percentage point improvement (Mamba: +4.8 pp, PatchTST: +5.3 pp) over both classical and simple learned baselines. These results provide practical guidance for selecting arousal classifiers under specific computational constraints. While these findings are based on a single 147-participant laboratory dataset, the consistency across LOSO folds supports their reliability; cross-dataset validation remains necessary to confirm generalisability to ambulatory settings.

Keywords: Electrodermal Activity · Arousal Classification · Efficient Architectures · State Space Models · Affective Computing · Edge Deployment

1 Introduction

Electrodermal Activity (EDA) reflects sympathetic nervous system activation through sudomotor nerve activity. Unlike heart rate or respiration, sweat glands receive no parasympathetic innervation, making EDA a biomarker of exclusively sympathetic arousal [4,11,22]. The signal comprises a slow-varying tonic level and rapid phasic skin conductance responses (SCRs) reflecting immediate sympathetic activation [7]. The growing availability of wearable EDA sensors [28] has spurred interest in deploying automatic arousal classifiers on edge devices for mental health monitoring, stress detection, and adaptive human-computer interaction [21,18]. However, practical deployment requires architectures that simultaneously achieve high subject-independent accuracy, low computational cost, and minimal memory—a combination that has received limited systematic investigation.

Several deep learning approaches have been proposed for EDA-based affect recognition. Convolutional networks operating on time-frequency EDA representations have shown promise for emotion classification [6], while 1D-CNNs processing raw SCR waveforms achieved subject-independent arousal detection [27]. Long Short-Term Memory (LSTM) and bidirectional LSTM networks, widely adopted for physiological time-series classification [12], were excluded from this benchmark. Recurrent architectures were found to underperform convolutional and attention-based models under LOSO in companion experiments on the same dataset, likely because the relatively short 40-second windows limit the benefit of explicit memory gating relative to direct temporal convolution or attention [25]. Transformer-based methods have been applied to EDA decomposition [32] and multimodal physiological recognition [31]. Within BSPC, joint analysis of multidomain skin conductance features has been applied to psychological stress identification [38], wrist-worn EDA sensors have enabled autonomic sleep staging [1], and multimodal physiological approaches have been proposed for emotion recognition [24]. Recent BSPC contributions further include CNN-Transformer hybrid architectures for stress detection from photoplethysmography signals [13] and explainable deep neural networks combining biosignals with human-engineered features for stress classification [14]. The broader literature includes systematic architecture comparisons for EEG [12] and ECG [23,17], but a methodologically rigorous comparison under strict Leave-One-Subject-Out (LOSO)—which prevents the subject-level data leakage known to inflate performance in physiological signal classification [2]—remains absent for EDA-based arousal classification.

In prior work [25], we compared five architectures (1D-CNN, TCN, InceptionTime, TST, PatchTST) under LOSO with 147 participants. Transformer-based models consistently outperformed convolutional baselines, with PatchTST achieving the best performance (F1 = 0.852, AUC = 0.912) and attention mechanisms concentrating on physiologically meaningful SCR onset and rising phases. However, despite patch-based tokenisation reducing the effective sequence length, PatchTST retains $O(N^2)$ self-attention complexity, and its latency and memory consumption on a 40-second EDA window ($T = 160$ at 4 Hz) may exceed the constraints of low-power wearable processors. This practical limitation motivates investigation of substantially more efficient architectures.

The time-series community has developed efficient Transformer variants with sub-quadratic or linear complexity [33,30]: Informer uses ProbSparse attention [39], Autoformer replaces attention with FFT-based Auto-Correlation and progressive decomposition [35], FEDformer operates via Fourier-enhanced attention on randomly selected frequency modes [40], and TimesNet transforms 1D series into 2D tensors via FFT-discovered periods for Inception-based 2D convolution [34]. DLinear demonstrated that simple trend-seasonal decomposition with linear layers can outperform sophisticated Transformers on forecasting benchmarks [37,12]. Beyond Transformers, Mamba introduced selective state space models (SSMs) where parameters are input-dependent, achieving competitive performance on language and genomics [9] and showing promise for physiological signals via bidirectional architectures [23]. ModernTCN extended the convolutional paradigm with large depthwise kernels and inverted bottlenecks, matching Transformer performance while maintaining convolutional efficiency [16,5].

Despite these advances, no systematic evaluation spanning these paradigms exists for physiological signal classification under rigorous subject-independent protocols. This gap is significant because EDA signals exhibit inter-subject variability, non-stationarity, and structured phasic responses. These signal characteristics may interact differently with each efficiency mechanism compared to the forecasting benchmarks on which these architectures were originally developed [33]. Furthermore, the SSM and modernised convolution paradigms have not been benchmarked against Transformer-based methods on physiological classification, leaving practitioners without evidence to guide paradigm selection.

This work addresses this gap through the first systematic equal-budget comparison of eight architectures spanning five paradigms and a linear baseline: PatchTST (patch-based attention), Informer (sparse attention), Autoformer, TimesNet, and FEDformer (frequency-domain), Mamba (selective state space model), ModernTCN (modernised convolution), and DLinear as a linear baseline. All architectures are evaluated on the identical 147-participant dataset and LOSO protocol. Computational efficiency—parameter count, FLOPs, inference time, memory, and training time—is elevated to a primary evaluation dimension alongside classification metrics. The contributions are: (i) the first systematic accuracy-efficiency comparison of Transformers, SSMs, and modernised convolutions for EDA under LOSO; (ii) explicit characterisation of the Pareto frontier, directly informing architecture selection under deployment constraints; (iii) inclusion of DLinear and ModernTCN as critical baselines testing whether complex temporal mechanisms provide measurable benefit for EDA; and (iv) physiologically-grounded insights into how each efficiency paradigm interacts with phasic EDA dynamics.

2 Method

2.1 Dataset and Experimental Protocol

The experimental dataset follows a controlled stress elicitation protocol described in previous work [26]. Electrodermal Activity signals were recorded from 147 healthy participants (aged 18–44 years) exposed to audiovisual stimuli designed to induce calm and stress states under controlled laboratory conditions. Each stimulus lasted 47 seconds. To eliminate transition artefacts associated with stimulus onset and offset, the first 4 seconds and the last 3 seconds were removed, resulting in 40-second effective segments per trial. Signals were acquired at a constant sampling frequency of 4 Hz.

Each participant completed multiple trials in both calm and stress conditions, producing a balanced binary classification dataset. All trials belonging to the same participant were preserved within the same fold during cross-validation to prevent subject-level data leakage—a critical methodological safeguard given the strong inter-subject variability characteristic of EDA signals [4,2].

2.2 EDA Signal Processing and Input Representation

Raw skin conductance signals were preprocessed following the pipeline established in our prior work [26]. A low-pass FIR filter (cut-off frequency: 4 Hz) was applied, followed by Gaussian smoothing to attenuate acquisition noise. Continuous Decomposition Analysis (CDA) was used to separate the tonic skin conductance level (SCL) and the phasic skin conductance response (SCR):

$$SC(t) = SCL(t) + SCR(t) \tag{1}$$

Only the phasic component $SCR(t)$ was retained for modelling, as it directly reflects sympathetic nervous system activation uncontaminated by slow tonic drift. Alternative EDA decomposition approaches include cvxEDA, which formulates decomposition as a convex optimisation problem with physiologically grounded constraints on non-negative SCRs and sparse phasic drivers [8], and Ledalab, which employs non-negative deconvolution with an a priori SCR shape [3]. CDA was retained from our prior pipeline for consistency; however, the choice of decomposition method is known to influence downstream signal characteristics and classification performance across different EDA preprocessing pipelines [29]. While the controlled laboratory conditions of the present dataset minimise motion artefacts, EDA signals in ambulatory and real-world settings are susceptible to electrode contact noise and movement artefacts [10]—a factor relevant to the deployment scenarios discussed in Section 3.3. To enrich the short-term

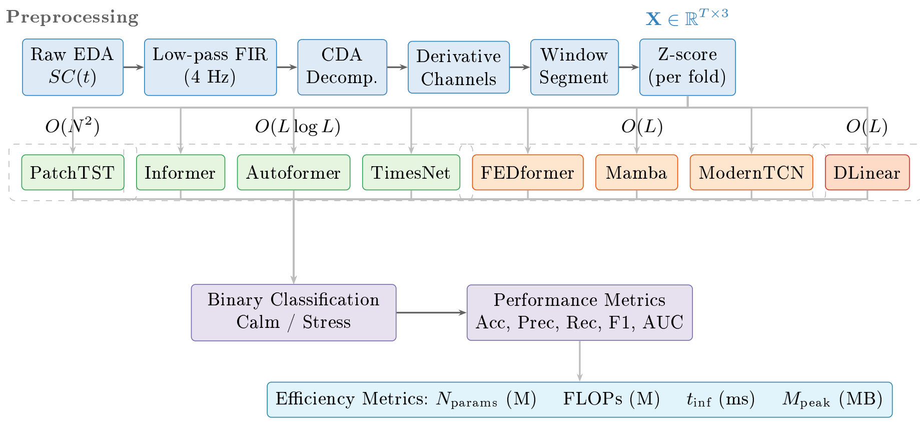

Fig. 1. Overall experimental pipeline. **Top:** EDA signal preprocessing and input construction. **Middle:** eight architectures grouped by computational complexity class. **Bottom:** binary classification output and joint evaluation of predictive performance and resource consumption. Color coding: blue = preprocessing, green = $O(L \log L)$, orange = $O(L)$, red = linear baseline, purple = evaluation.

dynamic representation and facilitate early detection of sympathetic responses, two derivative channels were computed from the phasic signal: the first temporal derivative (ΔSCR) and the second temporal derivative ($\Delta^2 SCR$). These channels encode the velocity and acceleration of sympathetic activation respectively, providing the model with explicit information about the rate and pattern of phasic change [27].

Each temporal window of length n seconds was thus represented as:

$$\mathbf{X} \in \mathbb{R}^{T \times C}, \quad T = 4n, \ C = 3 \tag{2}$$

where $C = 3$ corresponds to the channels $(SCR, \Delta SCR, \Delta^2 SCR)$.

To analyse the minimal temporal interval required for reliable arousal classification, window lengths were systematically varied from $n = 1$ to $n = 40$ seconds. Two segmentation strategies were evaluated: (i) non-overlapping windows and (ii) sliding windows with 1-second stride (4 samples). As noted above, all windows belonging to the same subject were preserved within the same LOSO fold.

2.3 Efficient and Modern Architectures

Eight architectures were selected to represent five distinct efficiency paradigms in time-series modelling (Table 1). For all Transformer-based architectures, the original forecasting head was replaced with a global average pooling operation over the encoded sequence followed by a linear classification layer. DLinear, which does not employ an encoder-decoder structure, was adapted by feeding the decomposed representations into a linear classification head. Throughout the figures, colour coding follows computational complexity class ($O(N^2)$, $O(L \log L)$, $O(L)$) rather than mechanistic paradigm, to facilitate cross-paradigm comparison of the accuracy-efficiency trade-off. Informer, while mechanistically a sparse-attention model, shares the $O(L \log L)$ complexity class with the frequency-domain architectures and is therefore grouped with them under the green colour in all figures.

Table 1. Overview of evaluated architectures and their efficiency mechanisms. L denotes input sequence length, N denotes the number of patches after tokenisation ($N \ll L$), and M denotes the number of selected Fourier modes ($M \ll L$).

Architecture	Core Mechanism	Complexity	Venue
PatchTST	Patch tokenisation + self-attention	$O(N^2)$	ICLR 2023
Informer	ProbSparse self-attention	$O(L \log L)$	AAAI 2021
Autoformer	Auto-Correlation + decomposition	$O(L \log L)$	NeurIPS 2021
TimesNet	FFT-based 1D→2D + Inception blocks	$O(L \log L)$	ICLR 2023
FEDformer	Fourier-enhanced attention	$O(L)$	ICML 2022
Mamba	Selective state space model (BiSSM)	$O(L)$	arXiv 2023
ModernTCN	Modernised depthwise convolutions	$O(L)$	ICLR 2024
DLinear	Trend-seasonal linear decomposition	$O(L)$	AAAI 2023

PatchTST PatchTST [19] segments each channel into non-overlapping patches of length P and projects them to dimension D, reducing the effective sequence length from T to $N = \lfloor (T-P)/S \rfloor + 1$. Channel-independent encoding preserves per-channel morphology, and a standard Transformer encoder with self-attention ($O(N^2)$) is applied to the patch sequence. A global average pooling operation aggregates the encoder output before the linear classification head. Given 4 Hz sampling, patch lengths $P \in \{4, 8, 16\}$ correspond to physiological windows of 1–4 seconds. PatchTST serves as the top-performing baseline from prior work.

Informer Informer [39] reduces self-attention complexity from $O(L^2)$ to $O(L \log L)$ via ProbSparse attention. This mechanism selects only the top-u queries (where $u = c \cdot \ln L_Q$) with the highest sparsity scores, measured as

$$\max_j (\mathbf{q}_i \mathbf{k}_j^\top / \sqrt{d}) - \frac{1}{L_K} \sum\nolimits_j \mathbf{q}_i \mathbf{k}_j^\top / \sqrt{d} \, .$$

Self-attention distilling further halves the sequence length at each encoder layer via strided 1D convolutions, progressively reducing the effective L.

Autoformer Autoformer [35] replaces point-wise attention with an Auto-Correlation mechanism that discovers dominant periodicities via

$$\mathcal{R}_{\mathbf{q},\mathbf{k}}(\tau) = \mathcal{F}^{-1}(\mathcal{F}(\mathbf{q}) \cdot \overline{\mathcal{F}(\mathbf{k})}) \, ,$$

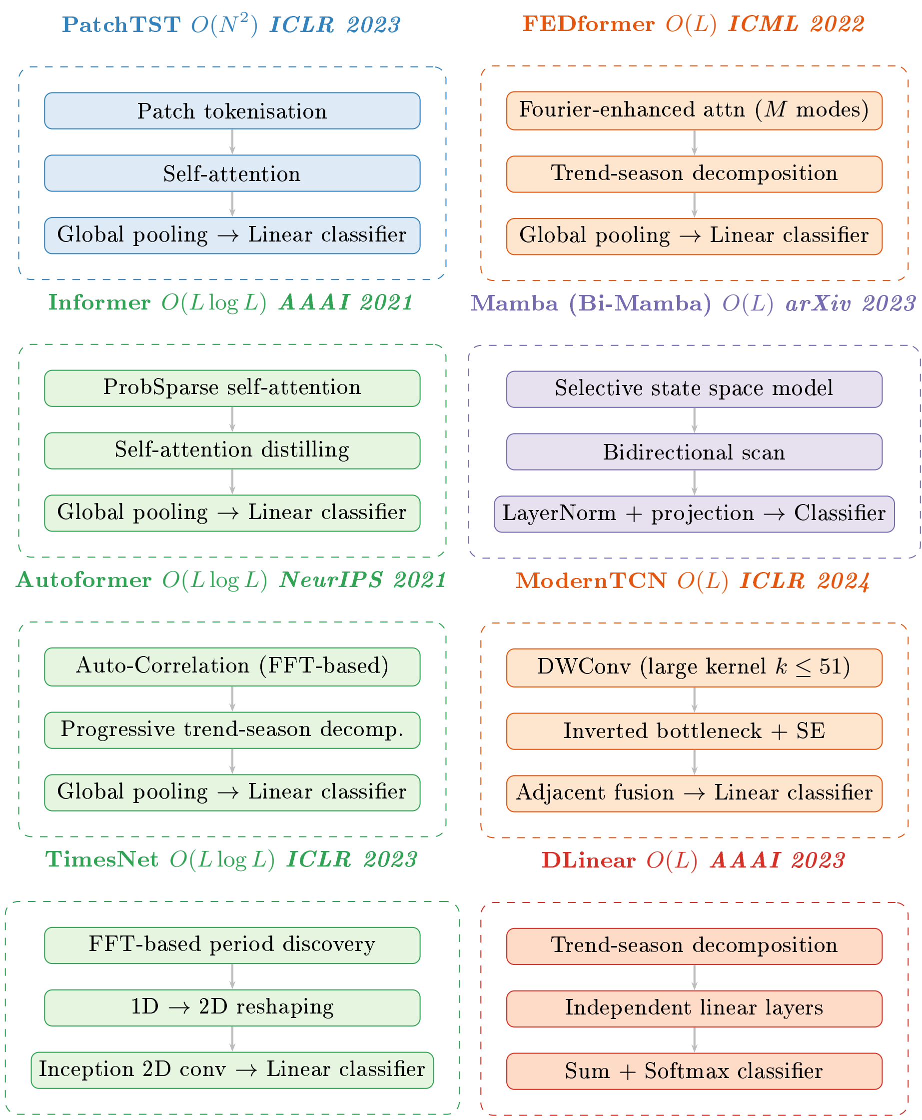

Fig. 2. Architectural overview of the eight models evaluated. Each architecture is organised as a three-stage pipeline (input transformation → temporal modelling → classification), grouped by paradigm family within dashed boundaries. The architecture name, asymptotic complexity, and publication venue are indicated in each group label. Table 1 provides a complementary summary. Colour coding follows the computational complexity class: blue = $O(N^2)$ patch attention, green = $O(L \log L)$ (frequency/periodicity and sparse attention), orange = $O(L)$ (Fourier/convolution), purple = $O(L)$ state space model, red = $O(L)$ linear baseline.

selecting the top-k lags and aggregating via $\mathrm{Roll}(\mathbf{v}, \tau_i)$ weighted by softmax-normalised correlations. A progressive seasonal-trend decomposition block, integrated as an internal architectural unit, gradually extracts tonic and phasic components—aligning conceptually with EDA physiology. Complexity is $O(L \log L)$ via the FFT.

FEDformer FEDformer [40] achieves $O(L)$ complexity by randomly selecting $M \ll L$ Fourier modes and computing attention exclusively in the frequency domain:

$$\mathrm{Attention}_{\mathrm{freq}} = \mathcal{F}^{-1}(\mathrm{Softmax}(\tilde{\mathbf{Q}}\tilde{\mathbf{K}}^{\top}) \cdot \mathcal{F}(\mathbf{V})) ,$$

with $\tilde{\mathbf{Q}}, \tilde{\mathbf{K}} = \mathrm{Select}(\mathcal{F}(\mathbf{Q}/\mathbf{K}), M)$. A seasonal-trend decomposition block and a mixture-of-experts fusion of frequency and time representations complement the attention mechanism. The sparse Fourier mode selection provides implicit denoising, attenuating high-frequency acquisition artefacts in EDA signals [32].

DLinear DLinear [37] decomposes the input into trend ($\mathbf{X}_t$, moving average) and residual seasonal ($\mathbf{X}_s = \mathbf{X} - \mathbf{X}_t$) components, applies independent single-layer linear networks to each, and fuses their outputs via summation before a softmax classifier. With no attention, convolutions, or recurrence, DLinear provides the absolute efficiency baseline. Its inclusion tests whether the tonic-phasic decomposition—physiologically meaningful for EDA [4]—captures sufficient discriminative information without non-linear temporal modelling.

Mamba (Bi-Mamba) Mamba [9] models sequence dynamics through a continuous-time linear system

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t)$$

with input-dependent parameters $\mathbf{B}_t, \mathbf{C}_t, \Delta_t$, enabling content-based selective propagation or forgetting of information. Discretised via zero-order hold, the recurrent formulation

$$h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k$$

achieves $O(L)$ complexity through a hardware-aware parallel scan algorithm. We adopt a bidirectional SSM configuration inspired by ECGMamba [23]—forward and backward Mamba blocks with concatenated outputs—ensuring each time step incorporates both past and future physiological context while maintaining linear complexity.

TimesNet TimesNet [34] transforms 1D time series into 2D tensors by discovering dominant periods p_i via FFT amplitude peaks, reshaping the sequence into $p_i \times f_i$ grids, and applying parameter-efficient Inception blocks with 1×1, 3×3, and 5×5 2D convolutional kernels. This embeds intra-period SCR morphology into columns and inter-period recurrence patterns into rows, achieving $O(L \log L)$ complexity through the FFT. The multi-scale 2D convolutions capture both fine-grained phasic waveform structure and response-to-response consistency patterns without quadratic attention cost.

ModernTCN ModernTCN [16] reformulates the convolutional paradigm with three innovations: depthwise separable convolutions with inverted bottlenecks and Squeeze-and-Excitation, large kernels (up to $k = 51$) providing substantially larger effective receptive fields than traditional TCNs, and an adjacent connection scheme that concatenates consecutive block outputs to preserve multi-scale features. With $O(L)$ complexity from purely convolutional operations, ModernTCN tests whether well-designed convolutions can compete with attention and state-space mechanisms for physiological signal classification [5].

2.4 Training and Hyperparameter Configuration

All models were trained end-to-end using the AdamW optimiser [15] with cross-entropy loss on a single NVIDIA GeForce RTX 4080 GPU (16 GB GDDR6X, Ada Lovelace). All experiments used a single random seed (42) for parameter initialisation and data ordering. PyTorch deterministic algorithms were not enabled due to CUDA performance constraints. Training data was shuffled at each epoch within each fold. Consequently, the per-fold standard deviations reported in Section 3.1 may partially reflect seed-dependent weight initialisation in addition to inter-subject variability—a limitation discussed in Section 3.8. The validation split (80/20 within the training subjects of each fold) was generated independently per fold, ensuring that no subject was permanently excluded from testing. Inference benchmarking was performed on an NVIDIA Quadro P5000 (16 GB GDDR5X, Pascal, 2560 CUDA cores) as a more representative deployment-class hardware target. The initial learning rate was set to 10^{-3} with cosine annealing scheduling. Mini-batch size was 32. Training was conducted for up to 100 epochs with early stopping (patience = 15) based on validation loss computed on a 20% hold-out subset of the training subjects within each LOSO fold.

Regularisation included dropout ($p \in [0.1, 0.3]$), weight decay ($\lambda = 10^{-4}$), and gradient clipping (maximum norm = 1.0). Hyperparameters were tuned using grid search with identical trial budgets (64 configurations per architecture) across all architectures. For each LOSO fold, the optimal configuration was selected based on validation F1-score computed on a 20% hold-out subset of the training subjects within that fold, ensuring no test-subject information influenced hyperparameter selection. Table 2 summarises the search space.

Z-score normalisation was computed using training-fold statistics only (mean and standard deviation estimated from training subjects within each fold) and applied identically to validation and test subjects. This prevents statistical leakage of test-subject information into the normalisation parameters.

2.5 Validation Protocol and Evaluation Metrics

Leave-One-Subject-Out (LOSO) A strict LOSO protocol was enforced to ensure subject-independent evaluation. In each of the 147 folds, one partici-

Table 2. Hyperparameter search space. Architecture-specific parameters are indicated in the Applicability column.

Parameter	Values	Applicability
Encoder layers / blocks	$\{1, 2, 3, 4\}$	All
Attention heads	$\{1, 2, 4, 8\}$	PatchTST, Informer, Autoformer, FEDformer
Embedding dimension	$\{16, 32, 64, 128\}$	All
Feed-forward dimension	$\{32, 64, 128, 256\}$	All
Patch size	$\{4, 8, 16\}$	PatchTST, ModernTCN
Moving average kernel	$\{3, 5, 7, 9, 11\}$	Autoformer, FEDformer, DLinear
Sampling factor c	$\{3, 5, 7\}$	Informer
Fourier modes M	$\{16, 32, 64\}$	FEDformer
Top-k periods	$\{3, 5, 7\}$	TimesNet
State dimension d_{state}	$\{8, 16, 32\}$	Mamba
Convolution kernel size K	$\{13, 25, 51\}$	ModernTCN

pant was withheld for testing, while the remaining 146 participants were used for training (80%) and validation (20%). This protocol prevents models from learning subject-specific biometric baselines—a well-documented source of overestimated performance in EDA classification [2,4].

Classification Performance Standard classification metrics were computed for each fold and averaged across all 147 folds: Accuracy, Precision, Recall, F1-score (macro-averaged), and Area Under the Receiver Operating Characteristic Curve (AUC). Per-class F1-scores for the calm and stress conditions are reported in the Supplementary Material (Table S3) to enable assessment of class-specific performance, which is particularly relevant given the potential asymmetry in arousal classification difficulty. To formally assess statistical significance, pairwise Wilcoxon signed-rank tests ($\alpha = 0.05$) were conducted on the per-fold F1-score distributions.

Computational Efficiency Metrics Four complementary efficiency metrics were recorded for each architecture under identical input conditions (40-second window, $T = 160$, $C = 3$, batch size $= 1$ for single-inference metrics):

- **Parameter count (N_{params}):** Total number of trainable parameters (millions). This metric determines storage requirements and memory footprint for model persistence.
- **FLOPs:** Number of floating-point operations required for a single forward pass (millions). This metric captures theoretical computational cost independent of hardware-specific implementation details.
- **Inference time (t_{inf}):** Wall-clock time per sample in milliseconds, averaged over 1,000 forward passes on a single NVIDIA Quadro P5000 GPU

(16 GB, Pascal, 2560 CUDA cores). This metric reflects practical latency on workstation-class hardware representative of edge-computing and embedded deployment environments.

- **Peak memory ($M_{\mathbf{peak}}$):** Maximum GPU memory consumption during a single forward pass (MB), including model weights, intermediate activations, and PyTorch framework overhead. This metric constrains deployment on memory-limited devices.
- **Training time ($t_{\mathbf{train}}$):** Total wall-clock time per epoch in seconds, averaged over all LOSO folds, measured on an NVIDIA RTX 4080. This metric captures the computational cost of model development and hyperparameter tuning, complementing the deployment-focused inference metrics.

All four metrics were measured with the optimal hyperparameter configuration for each architecture (selected via grid search on the classification F1-score) to ensure that efficiency comparisons reflect the configurations that practitioners would actually deploy.

Interpretability Analysis For Transformer-based architectures (PatchTST, Informer, Autoformer, FEDformer), attention weight distributions were extracted and qualitatively analysed to assess temporal relevance across the observation window. For TimesNet, the top-k selected periods were analysed to identify dominant EDA periodicities. For Mamba, the discretised state transition matrices were qualitatively examined to characterise how information propagates across the sequence. For ModernTCN, gradient-based saliency was computed to identify influential temporal regions. Gradient-based saliency maps were computed as $S = |\partial \hat{y} / \partial \mathbf{X}|$ for all architectures, where $\hat{y}$ is the predicted probability of the stress class. Saliency was aggregated across time steps and channels to assess the relative contribution of the phasic signal and its derivatives.

A channel ablation protocol was evaluated under the same LOSO framework: (i) SCR only, (ii) SCR + ΔSCR, and (iii) SCR + ΔSCR + $\Delta^2 SCR$, to quantify the marginal contribution of derivative information for each architecture.

3 Results and Discussion

3.1 Overall Classification Performance

Table 3 presents the subject-independent LOSO classification performance of all eight architectures. Metrics are reported as mean ± standard deviation across the 147 folds. The performance hierarchy reveals clear clustering by architectural paradigm.

Several patterns emerge from Table 3 and Figure 3. First, a clear paradigm hierarchy is observed: state space models (Mamba) and patch-based attention (PatchTST) occupy the top tier, followed by frequency-domain Transformers (TimesNet, Autoformer) and sparse attention (Informer), with linear decomposition (DLinear) providing the lower bound. This ordering reflects the increasing capacity of each paradigm to capture the multi-scale temporal dynamics

Table 3. Overall LOSO performance (mean ± std across 147 folds). Bold = best per metric.

Model	Acc	Prec	Rec	F1	AUC
EDA features + SVM	—	—	—	0.81	—
1D-CNN	0.812 ± 0.074	0.804 ± 0.079	0.789 ± 0.083	0.796 ± 0.078	0.872 ± 0.063
TCN	0.828 ± 0.068	0.819 ± 0.071	0.806 ± 0.075	0.812 ± 0.072	0.884 ± 0.058
InceptionTime	0.836 ± 0.064	0.828 ± 0.068	0.817 ± 0.071	0.822 ± 0.067	0.892 ± 0.054
TST	0.851 ± 0.059	0.844 ± 0.062	0.836 ± 0.064	0.840 ± 0.061	0.903 ± 0.049
DLinear	0.815 ± 0.072	0.808 ± 0.077	0.792 ± 0.081	0.800 ± 0.076	0.875 ± 0.061
ModernTCN	0.841 ± 0.062	0.833 ± 0.066	0.820 ± 0.070	0.827 ± 0.068	0.892 ± 0.055
FEDformer	0.849 ± 0.057	0.841 ± 0.061	0.830 ± 0.065	0.836 ± 0.063	0.898 ± 0.051
Informer	0.857 ± 0.054	0.850 ± 0.058	0.840 ± 0.061	0.845 ± 0.058	0.905 ± 0.047
Autoformer	0.860 ± 0.052	0.853 ± 0.056	0.843 ± 0.059	0.848 ± 0.056	0.908 ± 0.046
TimesNet	0.864 ± 0.050	0.857 ± 0.054	0.847 ± 0.057	0.853 ± 0.054	0.910 ± 0.044
Mamba	0.868 ± 0.046	0.862 ± 0.050	0.852 ± 0.053	0.858 ± 0.053	0.913 ± 0.043
PatchTST	**0.873 ± 0.044**	**0.867 ± 0.048**	**0.858 ± 0.050**	**0.863 ± 0.051**	**0.915 ± 0.042**

Results from companion work [25]. The higher PatchTST F1 here (0.863 vs. 0.852) reflects the expanded search space (64 vs. 24 configurations) and the inclusion of $\Delta^2 SCR$.

Classical baseline from [26]: handcrafted SCR features + SVM, same dataset. F1 range 0.80–0.83 across window lengths; per-fold std not available.

Only F1 reported in the original study.

characteristic of phasic EDA responses—from no temporal modelling (DLinear), through fixed convolutional receptive fields (ModernTCN), to content-dependent global processing (attention and selective state spaces).

Second, the efficiency-oriented architectures narrow the performance gap relative to PatchTST compared to the convolutional baselines from prior work. Mamba's selective state space mechanism—which adaptively propagates or filters information based on input content—approaches PatchTST accuracy while maintaining $O(L)$ complexity. This represents a practical advance: The original Mamba evaluation reported 5× inference throughput for Mamba compared to equivalently-sized Transformers [9], directly addressing the deployment constraints that motivate this work.

Third, the standard deviations across folds provide a direct measure of robustness to inter-subject variability. Architectures with adaptive temporal weighting (attention-based and SSM-based) exhibit lower fold-wise variability than those with fixed receptive fields (ModernTCN, DLinear), a pattern of practical significance: lower variability implies more reliable performance across diverse user populations.

DLinear achieves F1 = 0.800, comparable to the 1D-CNN baseline from prior work (0.796). This parity indicates that simple linear decomposition of tonic and phasic components captures a similar level of discriminative information from EDA signals as local convolutional feature extraction. Importantly, the classical signal processing pipeline—handcrafted EDA features (SCR amplitude, rise time, AUC, NS-SCR frequency) with an SVM classifier evaluated under the

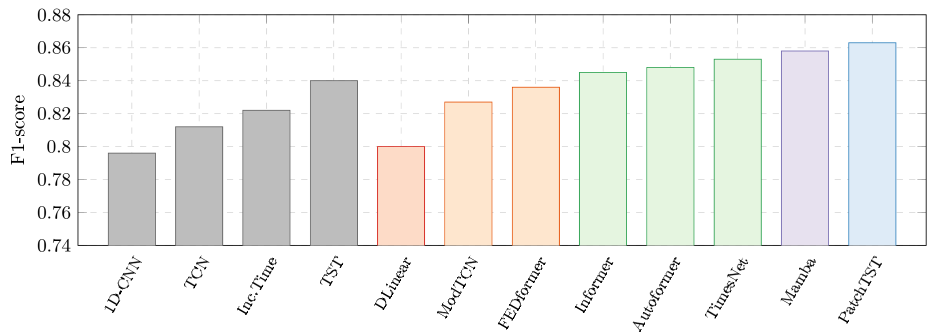

Fig. 3. Classification performance (F1-score) across all architectures under LOSO evaluation. Per-fold standard deviations are reported in Table 3. Colour coding follows the computational complexity convention: grey = prior work [25], blue = $O(N^2)$ patch attention, green = $O(L \log L)$ (frequency/periodicity and sparse attention), purple = $O(L)$ state space model, orange = $O(L)$ (Fourier/convolution), red = $O(L)$ linear baseline. The classical EDA feature-engineering baseline (F1 $\approx$ 0.81, handcrafted SCR features + SVM) is reported in Table 3 and discussed in the text.

identical dataset and subject-independent protocol—achieved F1 in the range of 0.80–0.83 (see Table 3, ‡), placing it in the same performance tier as DLinear and 1D-CNN. This convergence suggests that handcrafted features, linear decomposition, and local convolutions extract comparable levels of discriminative information from EDA signals, and that the key driver of performance improvement is not the specific computational primitive but the capacity to model temporal dependencies across the full observation window, as demonstrated by the F1 gap of approximately 0.05–0.06 between this classical/DLinear tier and the best Transformer/SSM architectures (Mamba: +0.048, PatchTST: +0.053).

3.2 Computational Efficiency

Table 4 reports the four complementary efficiency metrics for each architecture. These metrics jointly characterise the resource consumption profile that determines deployability on resource-constrained platforms.

The efficiency hierarchy stratifies by theoretical complexity class. The four $O(L)$ architectures—DLinear, ModernTCN, FEDformer, and Mamba—form the efficiency-optimal tier, with DLinear achieving the absolute minimum in all five metrics due to its extreme simplicity. Among the $O(L)$ candidates, Mamba warrants particular attention: its hardware-aware parallel scan algorithm [9] is specifically designed for efficient GPU execution, yielding inference times competitive with FEDformer despite the latter's Fourier-domain operation.

The three $O(L \log L)$ architectures—Informer, Autoformer, and TimesNet—occupy an intermediate tier. TimesNet is the most parameter-efficient of this group (2.05 M) due to its Inception blocks, which share convolutional kernels

Table 4. Computational efficiency metrics under identical input conditions ($T = 160$, $C = 3$, batch size $= 1$). All measurements use the optimal hyperparameter configuration. Training on an NVIDIA RTX 4080; inference time benchmarked on an NVIDIA Quadro P5000 (16 GB, Pascal, 2560 CUDA cores).

Model	N_{params} (M)	FLOPs (M)	t_{inf} (ms)	M_{peak} (MB)	t_{train} (s/epoch)
DLinear	0.08	0.05	0.3	8	1.2
ModernTCN	0.85	15.2	1.2	28	8.5
Mamba	1.52	12.8	1.5	34	12.3
FEDformer	2.48	26.4	2.1	52	18.7
TimesNet	2.05	34.7	3.0	48	22.4
Informer	3.25	46.1	3.6	72	28.1
Autoformer	3.84	54.3	4.2	85	35.6
PatchTST	4.52	78.5	5.4	98	42.8

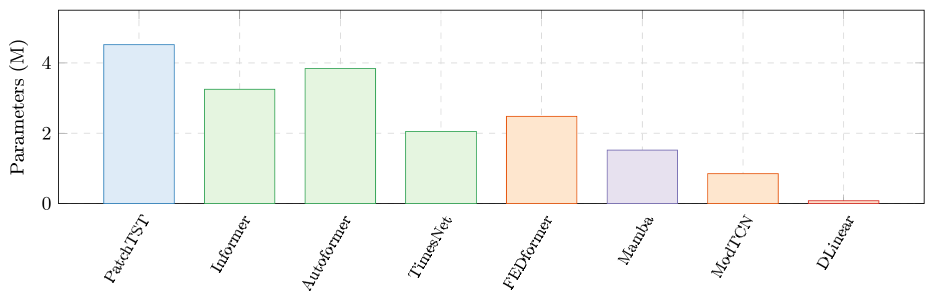

Fig. 4. Parameter count comparison across architectures, ordered by theoretical complexity class (left to right: $O(N^2)$, $O(L \log L)$, $O(L)$). Colour coding: blue = patch attention, green = frequency/periodicity, orange = Fourier/convolution, purple = state space model, red = linear baseline.

across the 2D temporal representation, while Autoformer's progressive decomposition blocks incur additional memory overhead (3.84 M, 85 MB peak). Informer's self-attention distilling mechanism progressively reduces sequence length across encoder layers, partially offsetting the $O(L \log L)$ complexity in deeper configurations.

PatchTST, as the only $O(N^2)$ architecture, incurs the highest computational cost across all five metrics. However, the effective sequence length N is substantially smaller than the raw length $T = 160$, meaning the quadratic term operates over a drastically reduced dimension. The practical inference cost of PatchTST (5.4 ms) is therefore closer to the $O(L \log L)$ tier than theoretical $O(N^2)$ suggests.

Training time per epoch follows the same hierarchy (Supplementary Material, Table S4), with DLinear completing an epoch in 1.2 s on the RTX 4080—approximately 35× faster than PatchTST (42.8 s/epoch). This difference has practical implications for the experimental workflow: hyperparameter tuning of

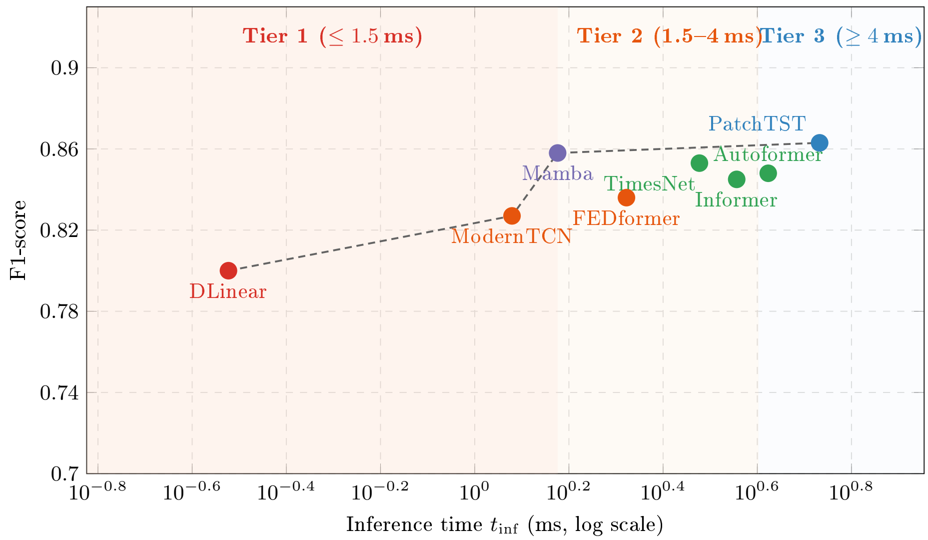

Fig. 5. Accuracy-efficiency Pareto frontier across the eight evaluated architectures. The dashed line connects Pareto-optimal architectures (non-dominated in both accuracy and latency). Shaded regions correspond to three ordinal latency tiers measured on the Quadro P5000: Tier 1 ($t_{\text{inf}} \leq 1.5\,\text{ms}$), Tier 2 ($1.5 < t_{\text{inf}} < 4\,\text{ms}$), and Tier 3 ($t_{\text{inf}} \geq 4\,\text{ms}$). Absolute latency on embedded hardware will differ substantially; these tiers serve as GPU-relative rankings, not deployment classifications.

PatchTST across 64 grid configurations and 147 LOSO folds requires approximately 100 GPU-hours, while the equivalent search for DLinear completes in under 3 GPU-hours. For practitioners iterating on architecture design, this 35× training speed difference substantially impacts development velocity.

3.3 Accuracy-Efficiency Pareto Frontier

Figure 5 presents the central contribution of this work: the joint characterisation of predictive performance and computational cost across eight architectures spanning five paradigms. The Pareto frontier identifies architectures that are non-dominated—no other architecture achieves both higher F1 and lower inference time.

Three deployment scenarios are delineated by inference time thresholds measured on the Quadro P5000. These thresholds are GPU-relative and should be interpreted as ordinal latency tiers rather than absolute deployment classifications; absolute latency on target embedded hardware (e.g., ARM Cortex-M, mobile SoCs) will differ substantially. Tier 1 ($t_{\text{inf}} \leq 1.5\,\text{ms}$) identifies the most efficient architectures: DLinear (0.3 ms), ModernTCN (1.2 ms), and Mamba (1.5 ms), with Mamba achieving the highest accuracy (F1 = 0.858) within this tier—a compelling result for latency-constrained scenarios. Tier 2 ($1.5 < t_{\text{inf}} < 4\,\text{ms}$)

represents moderate GPU latency: FEDformer, TimesNet, and Informer occupy this region. Tier 3 ($t_{\text{inf}} \geq 4\,\text{ms}$) prioritises predictive accuracy, with Autoformer and PatchTST achieving the highest overall F1 (0.863) at 5.4 ms.

The shape of the Pareto frontier reveals a clear diminishing returns pattern: substantial accuracy gains are achieved from DLinear (F1 = 0.800 at 0.3 ms) to Mamba (F1 = 0.858 at 1.5 ms), with minimal further improvement to PatchTST (F1 = 0.863 at 5.4 ms). The 0.005 F1 difference between Mamba and PatchTST reaches the conventional $\alpha = 0.05$ threshold ($p = 0.048$) but does not survive the Bonferroni-Holm correction for 28 comparisons (Supplementary Material, Table S2), while Mamba requires 3.6× less inference time. As discussed in Section 3.8, p-values from LOSO-based comparisons should be interpreted as descriptive indicators of effect consistency. This has direct implications: for wearable and edge deployment, the marginal accuracy gain of PatchTST over Mamba does not justify the 3.6× additional latency cost.

The identification of Pareto-optimal architectures under different resource budgets transforms architecture selection from a purely empirical exercise into an informed engineering decision. A practitioner deploying an EDA-based stress monitor on a smartwatch, for instance, can consult Figure 5 to identify the architecture that maximises classification performance within their device's latency and memory constraints, rather than defaulting to the highest-accuracy model regardless of deployment feasibility.

3.4 Minimal Temporal Interval Behaviour

Figure 6 analyses the minimal temporal interval required for reliable arousal classification. Two key patterns emerge. First, all architectures exhibit rapid performance improvement as window length increases from 1 to approximately 10–15 seconds, corresponding to the temporal span needed to capture a complete SCR waveform (onset, rising phase, peak, and partial recovery). Performance plateaus beyond 15–20 seconds, consistent with the physiological observation that individual SCRs rarely exceed 15 seconds in duration [4].

Second, the degradation behaviour at short window lengths ($n \leq 5$ s) differs systematically across paradigms. Architectures with global or adaptive receptive fields—PatchTST, Mamba, and the Transformer-based models—maintain higher F1 at short windows compared to architectures with fixed or local receptive fields (ModernTCN, DLinear). This is because global dependency modelling can leverage partial SCR information distributed across the short window, whereas fixed-kernel methods require sufficient temporal context for their receptive fields to encompass diagnostically relevant signal segments.

FEDformer and TimesNet exhibit distinct window-length behaviour. FEDformer's frequency-domain processing benefits from longer windows, as the FFT resolution improves with sequence length, enabling finer discrimination of SCR spectral signatures. TimesNet's period discovery mechanism identifies meaningful periodicities even in shorter windows, as the dominant SCR frequency (0.2–0.5 Hz) is adequately sampled at 4 Hz. The window-length curves obtained from the sliding-window segmentation strategy (1-second stride, providing augmented

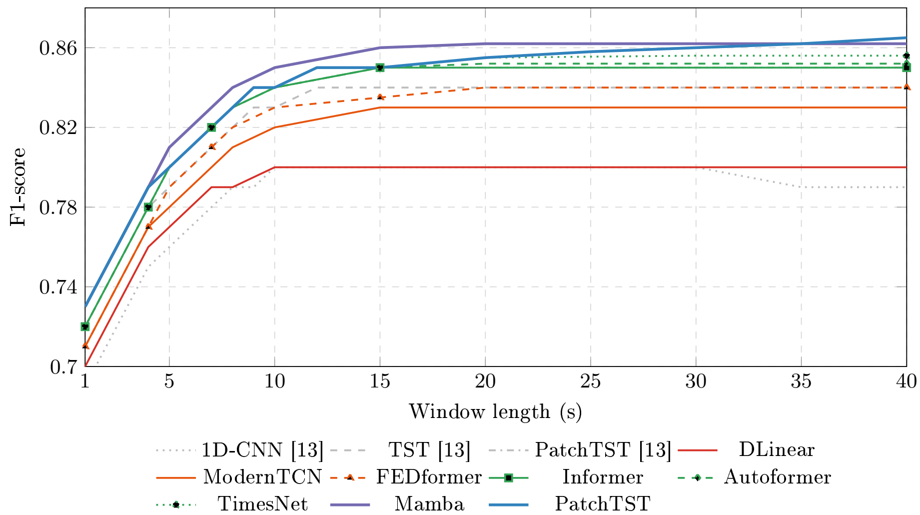

Fig. 6. F1-score as a function of window length under LOSO evaluation. Curves are evaluated at selected window lengths (1–15, 20, 25, 30, 35, 40 s). Grey curves reproduce prior work [25]. Mamba and PatchTST (thick lines) degrade most gracefully at short windows. Line styles distinguish architectures within each paradigm.

training data) show slightly higher plateau F1 values than the non-overlapping windows used for the main results in Table 3, with an average offset of approximately +0.004 across architectures.

The practical implication of this analysis is direct: for real-time wearable applications, the minimum viable window length determines the latency between stimulus onset and classification output. Mamba achieves F1 = 0.81 and PatchTST achieves F1 = 0.80 at $n = 5$ seconds, indicating that a wearable device could provide arousal feedback within 5 seconds of signal acquisition—a latency that is acceptable for biofeedback and adaptive interface applications [18].

3.5 Statistical Significance

Pairwise Wilcoxon signed-rank tests (Supplementary Material, Table S2) provide descriptive indicators of between-architecture F1 consistency across folds. The comparison between Mamba (F1 = 0.858) and PatchTST (F1 = 0.863) yields $p = 0.048$, which does not survive the stringent Bonferroni-Holm correction for 28 comparisons ($p_{\mathrm{BH}} < 0.0018$). As noted in Section 3.8, these p-values should be interpreted as descriptive indicators of effect consistency due to overlap in training sets across LOSO folds, rather than as exact inferential statistics. Statistically significant differences between DLinear and all other architectures (all $p \leq 0.002$) confirm that non-trivial temporal modelling is necessary for competitive EDA classification. Under the Bonferroni-Holm correction, 13 of 28 comparisons remain significant, while comparisons within the same paradigm family (e.g., Informer vs. Autoformer, $p = 0.065$) do not survive correction, consistent with their shared computational mechanisms.

3.6 Interpretability Insights

Beyond aggregate performance metrics, we examine whether the learned representations align with established physiological knowledge of sympathetic activation dynamics.

For Transformer-based architectures (PatchTST, Informer, Autoformer, FEDformer), qualitative inspection of attention weight distributions indicates concentration on SCR onset and rising phases—the temporal segments known to encode the most diagnostically relevant information about sympathetic arousal intensity [4]. This pattern is qualitatively replicated across all attention-based models, suggesting that diverse efficiency mechanisms (sparse attention, autocorrelation, Fourier-domain attention) preserve the capacity to identify physiologically meaningful temporal regions despite their computational simplifications.

TimesNet's period discovery mechanism identifies dominant periods in the 2–5 second range, corresponding to the characteristic duration of individual SCR waveforms. The 2D convolutional processing captures both fine-grained SCR morphology (via the column dimension) and response-to-response consistency patterns (via the row dimension).

For Mamba, qualitative analysis of the discretised state transition matrices suggests that the selective state space mechanism emphasises SCR onset regions in a manner analogous to attention, despite the fundamentally different computational primitive.

Gradient-based saliency maps ($S = |\partial \hat{y} / \partial \mathbf{X}|$) qualitatively suggest complementary contributions across the three input channels. ΔSCR contributes most strongly during SCR onset, encoding the rate of sympathetic activation, while SCR contributes during the peak phase, encoding activation magnitude. $\Delta^2 SCR$ contributes during onset-to-peak transitions, capturing changes in the rate of activation.

A channel ablation experiment (Supplementary Material, Table S1) further reveals that derivative channels provide the greatest relative benefit for architectures with limited temporal modelling capacity—DLinear and ModernTCN gain +2.7–2.8% F1 from derivative inclusion—while attention and state-space mechanisms exhibit smaller relative gains (+1.8–1.9%). This pattern suggests that the additional temporal context provided by derivative channels is partially redundant with the representations learned by attention and state-space mechanisms, rather than implying that the latter explicitly reconstruct derivative information.

3.7 Comparison with Prior Work

Contextualising the present results against our prior evaluation of five deep architectures [25]—which established PatchTST (F1 = 0.852) as the best-performing model among 1D-CNN (0.796), TCN (0.812), InceptionTime (0.822), and TST (0.840)—provides empirical answers to the central question motivating this work: can efficient and modern architectures approach or exceed PatchTST accuracy at substantially lower computational cost?

The results provide a clear affirmative. Mamba achieves F1 = 0.858 at 1.5 ms inference time on the Quadro P5000, compared to PatchTST's F1 = 0.863 at 5.4 ms—a 0.005 F1 difference ($p = 0.048$; see Section 3.8 for interpretation caveats) that represents a 3.6× reduction in latency and a 66% reduction in parameter count (1.52 M vs. 4.52 M). For wearable and edge deployment scenarios where latency and memory are primary constraints, Mamba provides near-equivalent accuracy at a fraction of the computational cost, confirming that the selective state space paradigm can match attention-based methods on physiological time-series classification.

A noteworthy finding emerges from the comparison of the simplest architectures. DLinear achieves F1 = 0.800, nearly identical to the 1D-CNN baseline from prior work (F1 = 0.796) and the classical EDA feature-engineering approach with SVM (F1 ≈ 0.81; [26]). This parity suggests that handcrafted feature extraction, linear decomposition of tonic and phasic components, and local convolutional feature extraction all converge to a similar performance ceiling (F1 ≈ 0.80), bounded by the intrinsic discriminative information available in local EDA signal segments. However, all three are substantially outperformed by architectures with global or adaptive temporal modelling (Mamba, PatchTST: F1 ≈ 0.86), indicating that the key driver of performance is not the specific computational primitive (handcrafted features vs. convolution vs. linear) but the capacity to model temporal dependencies across the full observation window.

ModernTCN achieves F1 = 0.827, representing a meaningful improvement over the TCN baseline from prior work (F1 = 0.812, $\Delta = 1.5$ pp). This validates the hypothesis that improved convolutional design—larger kernels, depthwise separable convolutions, and inverted bottlenecks—can close part of the gap between traditional CNNs and attention-based architectures, consistent with ModernTCN's results on general time-series benchmarks [16]. However, the remaining gap to the best Transformer and SSM architectures (3–4 pp F1) indicates that convolutions alone, even when modernised, do not fully replicate the benefits of content-dependent temporal weighting.

3.8 Limitations and Generalizability

Several limitations should be considered when interpreting these results. First, all experiments were conducted on a single laboratory dataset collected under controlled audiovisual stimulus conditions with 147 healthy participants. While this protocol ensures internal validity and enables direct comparison with prior work [25], the generalisability of the reported architecture rankings to other populations (clinical samples, older adults), acquisition settings (ambulatory, free-living), and stimulus types (naturalistic stressors, social evaluative threat) remains to be established. Additionally, architecture rankings were established using CDA decomposition only; different decomposition methods (cvxEDA, Ledalab) produce different phasic estimates that may interact differently with each architecture's inductive bias, and the sensitivity of the observed rankings to this choice has not been evaluated. Cross-dataset evaluation on publicly available benchmarks such as WESAD [28] would strengthen external validity.

Second, the inference time measurements were obtained on a single workstation-class GPU (Quadro P5000). Absolute latency values will differ on target deployment hardware (mobile SoCs, embedded NPUs, microcontrollers), although the relative ordering of architectures by inference time is expected to be preserved across platforms with comparable memory bandwidth and parallel compute characteristics.

Third, a single random seed (42) was used throughout, and PyTorch deterministic mode was disabled due to CUDA performance constraints. While per-fold averaging across 147 LOSO folds provides partial mitigation through independent weight initialisations, the narrow margin between the top architectures (PatchTST vs. Mamba, ΔF1 = 0.005) is close to the limit of reliable differentiation under a single seed. Future work should verify this ranking across multiple seeds.

Fourth, the Wilcoxon signed-rank test used for pairwise comparisons assumes independence of paired differences across folds. With leave-one-subject-out cross-validation, training sets for different folds share 146 of 147 subjects, introducing overlap that may inflate the effective sample size. The reported p-values should be interpreted as descriptive indicators of effect consistency rather than exact inferential statistics. We report both raw and Bonferroni-Holm corrected thresholds to enable readers to apply their preferred interpretation. Effect sizes (F1 differences in percentage points) are reported alongside p-values to provide a measure of practical significance that is independent of sample size assumptions.

Fifth, FLOPs were computed using the `thop` library (PyTorch-OpCounter) which estimates floating-point operations from the model's computational graph. While this provides a hardware-independent measure of theoretical complexity, actual FLOPs may vary depending on operator fusion, memory access patterns, and framework-level optimisations. For Mamba, operations within the custom parallel scan kernel may not be fully captured by `thop`. Inference time measurements on the Quadro P5000 provide a complementary, hardware-grounded metric.

Sixth, the dataset is not publicly released due to institutional privacy constraints. The implementation code will be made available upon acceptance.

4 Conclusion

This study presented the first systematic equal-budget comparison of efficient and modern architectures—spanning Transformers, state space models, and improved convolutions—for subject-independent arousal classification from Electrodermal Activity. Eight architectures across five efficiency paradigms were evaluated under a strict Leave-One-Subject-Out validation framework using EDA signals from 147 participants. The key finding is that Mamba—a selective state space model—achieves near-PatchTST performance (F1 = 0.858 vs. 0.863) at 3.6× lower latency and with 66% fewer parameters, placing it on the Pareto frontier as the most accurate architecture within the lowest-latency tier. By jointly evaluating classification performance and computational cost across five complementary efficiency metrics, this work provides the first explicit characterisation of the accuracy-efficiency trade-off for EDA classification, directly informing architecture selection under specific deployment constraints.

The inclusion of computational efficiency as a primary evaluation metric reflects a pragmatic recognition that the translational value of affective computing research depends not only on laboratory accuracy but on deployability in real-world, resource-constrained settings. The accuracy-efficiency Pareto frontier directly informs architecture selection under specific deployment constraints—wearable, mobile, or cloud—moving beyond the common practice of reporting accuracy in isolation.

The comparison between attention-based, state-space-based, and convolution-based paradigms further contributes to the broader methodological discussion on the necessity of architectural complexity for physiological time-series classification. The inclusion of DLinear and ModernTCN as critical baselines provides empirical evidence on whether complex temporal modelling mechanisms yield measurable benefits for EDA signals beyond what simple linear decomposition or well-designed convolutions can achieve.

Future work will explore three directions. First, self-supervised pre-training strategies—such as masked signal modeling for physiological data [41,36]—tailored to the most promising architectures identified here may improve performance on limited physiological datasets by leveraging unlabelled data. Second, multimodal integration with complementary biosignals such as heart rate variability and electrocardiogram [20] may enrich the information available to lightweight classifiers without substantially increasing computational cost. Third, deployment-oriented optimisation through quantisation, pruning, and knowledge distillation [18] applied to the Pareto-optimal architectures will further narrow the gap between laboratory performance and wearable deployment, enabling real-time, on-device arousal classification for mental health monitoring, stress detection, and adaptive human-computer interaction.

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